

Yejun (Henry) Kim

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Education

University of Virginia

Charlottesville, VA | 2021–2025

B.S. Aerospace Engineering & Mathematics

Cumulative GPA: 3.6/4.0

Honors / Exams: Dean's List, QuestBridge National College Match, AIME

Work Experience

Epic Systems | Technical Solutions Engineer

Madison, WI | Jul 2025–Present

- Wrote SQL search queries against customer production systems to surface performance and reliability metrics, resolving 20+ high-severity issues per month via root-cause investigation across 200+ databases
- Guided end-users through fixes and workarounds for production issues while escalating unresolved defects to R&D, contributing to 15 fixes / enhancements
- Built diagnostic tooling for recurring production issues, adopted by 2-3 implementation teams and reducing manual troubleshooting time by up to 40%

Traffic Operations Laboratory | Data Engineer

Charlottesville, VA | Spring 2025

- Built an ETL pipeline ingesting real-time OBD telemetry via Raspberry Pi, cleaning data to exclude idle periods and duplicates, feeding a human-in-the-loop connected cruise control research system

Scholar's Lab Makerspace | Software Developer & Technologist

Charlottesville, VA | 2022–2025

- Maintained and improved a Linux / Python photo booth system (camera, printer, monitor, Raspberry Pi) used by hundreds of students and visitors at public events, resolving 20 issues over its lifecycle
- Built a JavaScript macro to process and deduplicate visitor survey data into digestible reports

Projects & Activities

Underwater Robotics Team | Software Co-Lead

Charlottesville, VA | 2024–2025

- Built a stereo-vision system for an underwater ROV using dual cameras and OpenCV, integrating with the hardware team; developed recalibration procedures to maintain accuracy as camera positioning shifted during operation

Guitar Learning Platform

Summer 2025–Present

- Architected a full-stack guitar practice-tracking application with REST APIs for practice tracking, media uploads, and guitar marketplace scraping
- Developing ML pipelines (PyTorch, CNNs, YOLOv8, CQT/MFCC spectrograms) for computer-vision and audio classification of guitar practice sessions

Orbital Transfer Calculator

Dec 2025

- Developed a web-based orbital mechanics tool visualizing Hohmann, bi-elliptic Hohmann, common apse line, apse line rotation, and plane-change trajectories in 3D via the Desmos API
- Solved Lambert's problem across orbital planes and applied Nelder-Mead optimization to compute minimum delta-V plane-change trajectories

Skills

Languages: Python, JavaScript, SQL, Java, C++, MATLAB, R

Frameworks/Libraries: React, FastAPI, Flask, PyTorch, NumPy, pandas, SciPy, scikit-learn, OpenCV, Supabase

Developer Tools: Git, GitHub, Linux, Raspberry Pi, VS Code